

GLASGOW COLLEGE UESTC

Exam paper

Digital Signal Processing (UESTC4005)

Date: (remember to complete when info available from Ruoli)

Time: (remember to complete when info available from Ruoli)

Attempt all PARTS. Total 100 marks

**Use one answer sheet for each of the questions in this exam.
Show all work on the answer sheet.**

Make sure that your University of Glasgow and UESTC Student Identification Numbers are on all answer sheets.

An electronic calculator may be used provided that it does not allow text storage or display, or graphical display.

All graphs should be clearly labelled and sufficiently large so that all elements are easy to read.

The numbers in square brackets in the right-hand margin indicate the marks allotted to the part of the question against which the mark is shown. These marks are for guidance only.

Q1 Let $x_1(t)$ and $x_2(t)$ be continuous-time signals with spectrum given in Fig Q1:

Continued overleaf

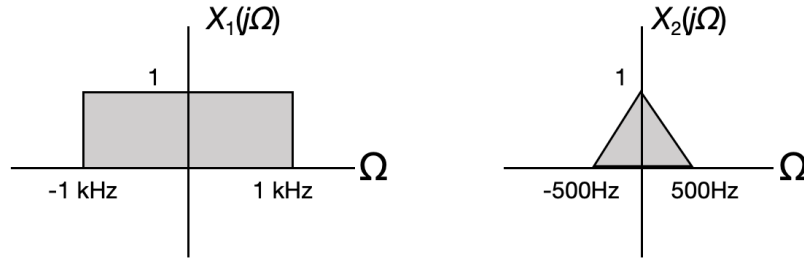


Fig Q1 The spectrum of two continuous-time signals $x_1(t)$ and $x_2(t)$

Let $x_1[n]$ and $x_2[n]$ be sampled discrete-time signals of $x_1(t)$ and $x_2(t)$, respectively, with sampling frequency $F_s=4\text{kHz}$. Denote a system $y[n]=x_1[n]+(-1)^n x_2[n]$. Answer the following questions:

- (1) Sketch $X_1(e^{j\omega})$ and $X_2(e^{j\omega})$; (5 points)
- (2) Sketch $Y(e^{j\omega})$; (5 points)
- (3) Design a system to recover $x_1(t)$ and $x_2(t)$ from $y[n]$. Specify your system in the form of a block diagram and justify your design. (5 points)

Q2 Consider a **causal** LTI system shown in Fig Q2, which consists of four subsystems.

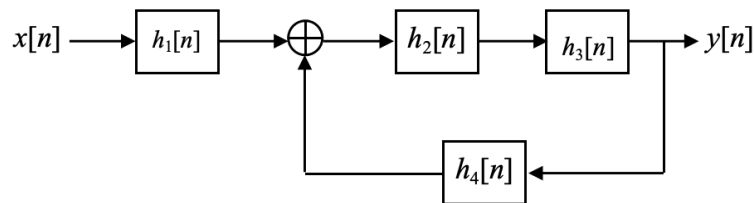


Fig Q2 A causal LTI system

In particular, the impulse responses of the first and the second subsystems are respectively given by

$$h_1[n]=2\delta[n-1]+\delta[n-2] \text{ and } h_2[n]=\delta[n-2]-\delta[n-1]$$

The third subsystem is characterized by the following linear difference equation:

$$y[n] = x[n] + x[n-1]$$

and the fourth subsystem is characterized as:

$$y[n] = -0.5y[n-1] + 2x[n]$$

Answer the following questions:

- (1) What is impulse the response $h_4[n]$ of the fourth subsystem? (5 points)
- (2) Which subsystem has the longest impulse response? Why? (5 points)
- (3) Let $h[n]$ denote the impulse response of the whole LTI system. Find the frequency domain response $H(e^{j\omega})$ of $h[n]$. (5 points)

Continued overleaf

Q3 A real signal $x[n]$ is passed through an LTI system with impulse response $h[n]$ and frequency response $H(e^{j\omega})$ to produce a real output $y[n]$. Fig Q3 are showing the magnitude and phase of $X(e^{j\omega})$ and $Y(e^{j\omega})$:

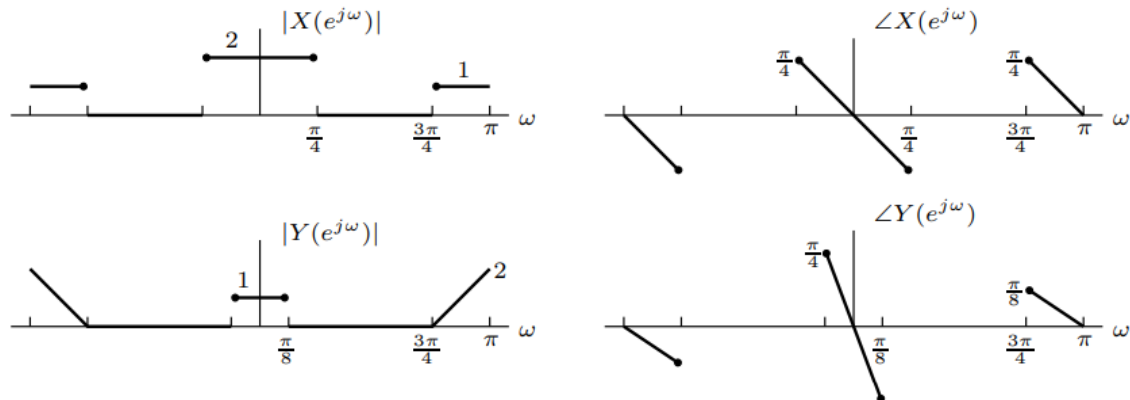


Fig Q3 The magnitude and phase of $X(e^{j\omega})$ and $Y(e^{j\omega})$

Answer the following questions:

- (1) Can we uniquely determine $H(e^{j\omega})$ based on the given information of $X(e^{j\omega})$ and $Y(e^{j\omega})$? Why? (5 points)
- (2) Sketch the magnitude and phase spectrum of $H(e^{j\omega})$. **Note:** you should also clearly label regions of $|H(e^{j\omega})|$ and $\angle H(e^{j\omega})$, which cannot be determined from the above given information, if your answer to the first question is No. (5 points)
- (3) System identification for an LTI system is the process of determining the impulse response $h[n]$ by analyzing the relationship between the input and output signals. Specifically, we can determine $h[n]$ by passing a test signal $x_{\text{test}}[n]$ through the system and inferring $h[n]$ from the corresponding output $y_{\text{test}}[n]$. Verify whether the following signals can be used for $x_{\text{test}}[n]$ and give your explanation. (10 points)
 - (a) $x_1[n] = \delta[n-1]$, (b) $x_2[n] = \frac{\sin(n\pi/5)}{n\pi/5}$, (c) $x_3[n] = e^{j\pi n/10}$

Q4 Let $H(z)$ be the transfer function of a causal LTI system

$$H(z) = \frac{1}{z^2 + 0.6z + 0.05}$$

- (1) Sketch the zero-pole plot of $H(z)$, and give its ROC. (4 points)
- (2) Give the difference equation of this discrete-time system. (3 points)
- (3) Compute the impulse response of the system, i.e., $h[n]$. (4 points)
- (4) Sketch the Direct form II structure of the system. (4 points)

Continued overleaf

Q5 Consider the following sequence

$$x[n] = \cos\left(\frac{5\pi n}{32}\right) + 2\cos\left(\frac{11\pi n}{32}\right), \quad 0 \leq n \leq 63$$

(1) Compute the 64-point DFT of $x[n]$. (5 points)

(2) Define $y_1[n]$ as

$$y_1[n] = \begin{cases} x[n], & 0 \leq n \leq 63 \\ x[n-64], & 64 \leq n \leq 127 \end{cases}$$

Please compute the 128-point DFT of $y_1[n]$. (5 points)

(3) Define $y_2[n]$ as

$$y_2[n] = \begin{cases} x[n/2], & n = 0, 2, 4, \dots, 126 \\ 0, & \text{others} \end{cases}$$

Please compute the 128-point DFT of $y_2[n]$. (5 points)

Q6 Please design a digital processing system to filter out the high-frequency noise components in a signal. It is known that the system's sampling rate is 20kHz, the spectral components of interest in the signal lie below 2.5kHz, and the noise components are above 5.5kHz. The requirements are that the response fluctuations in the frequency band of interest cannot exceed 0.5dB, and the noise suppression must be at least 15 dB. Please choose the filter type (FIR or IIR) by yourself and provide the detailed steps for the filter design.

(20 points)

End of question paper